

Derivative Rules Formula Sheet

A compact, readable reference for core derivative rules with rule-selection cues and examples.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/formula-sheet/derivative-rules/>

Name: _____ **Date:** _____

Linearity

$$\begin{aligned}(c)' &= 0 \\ (x^n)' &= nx^{n-1} \\ (f \pm g)' &= f' \pm g'\end{aligned}$$

Products and quotients

$$\begin{aligned}(fg)' &= f'g + fg' \\ (f/g)' &= (f'g - fg')/g^2\end{aligned}$$

Composition

$$(f \circ g)' = f'(g(x))g'(x)$$

Exponential and logarithmic

$$\begin{aligned}(e^x)' &= e^x \\ (a^x)' &= a^x \ln a \\ (\ln x)' &= 1/x\end{aligned}$$

Trigonometric

$$\begin{aligned}(\sin x)' &= \cos x \\ (\cos x)' &= -\sin x \\ (\tan x)' &= \sec^2 x\end{aligned}$$

Inverse trigonometric

$$\begin{aligned}(\arcsin x)' &= 1/\sqrt{1-x^2} \\ (\arctan x)' &= 1/(1+x^2)\end{aligned}$$